

LA GRANDE, OR, USA

WMO: 726884

Lat: 45.283N

Lon: 118.000W

Elev: 2713

StdP: 13.31

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24148

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.2	15.7	-1.7	5.5	9.6	6.5	8.5	17.5	34.7	37.5	31.8	35.0	5.0	320	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.1	93.4	63.2	90.7	62.4	88.0	61.5	65.4	88.8	63.8	87.2	62.3	84.8	8.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.7	75.7	72.5	54.5	69.8	71.0	52.5	64.8	69.9	31.5	88.4	30.3	87.0	29.2	84.7	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	24.0	20.6	-0.2	99.9	6.2	3.1	-4.7	102.2	-8.3	104.0	-11.8	105.8	-16.3	108.1		
(5)			-1.3	68.2	6.0	1.9	-5.7	69.6	-9.2	70.7	-12.6	71.8	-17.0	73.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	33.0	35.8	41.7	46.6	54.3	61.0	70.3	67.9	59.7	48.3	39.2	32.0		
	DBStd	14.66	7.92	7.65	6.45	6.22	6.97	6.74	5.63	5.61	6.92	7.22	8.51	8.23		
	HDD50	2394	528	399	266	139	37	4	0	0	7	118	337	559		
	HDD65	6133	992	817	721	553	336	155	19	37	187	517	775	1024		
	CDD50	2111	0	1	10	37	171	334	629	555	297	66	11	0		
	CDD65	376	0	0	0	0	5	35	183	126	27	0	0	0		
	CDH74	6957	0	0	0	21	173	684	2991	2285	760	43	0	0		
	CDH80	3065	0	0	0	2	35	227	1470	1049	274	8	0	0		
	WSAvg	8.2	11.3	9.8	8.7	8.2	7.3	6.9	6.5	6.2	5.9	7.2	9.7	11.2		
	PrecAvg	19.4	2.2	1.7	1.8	2.0	2.3	1.6	0.6	0.6	0.7	1.4	2.3	2.4		
(15) Precipitation	PrecMax	25.9	3.4	3.0	3.4	3.7	6.1	3.5	2.3	2.0	2.3	2.7	5.4	5.0		
	PrecMin	14.6	0.7	0.7	0.5	0.7	0.4	0.4	0.2	0.0	0.0	0.6	0.6	1.1		
	PrecStd	3.0	0.7	0.6	0.7	0.9	1.4	0.8	0.6	0.5	0.6	0.6	1.0	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	57.6	66.5	77.0	83.9	92.5	99.4	97.4	91.3	80.5	64.4	53.9		
		MCWB	42.8	46.0	49.4	55.8	60.1	65.0	65.3	63.6	61.2	56.8	49.8	45.0		
	2%	DB	46.4	53.5	62.9	71.7	79.1	86.0	95.0	93.1	87.8	72.4	60.8	48.1		
		MCWB	39.9	44.1	47.9	53.8	58.7	61.9	63.7	62.4	60.1	52.9	47.7	42.0		
	5%	DB	45.1	49.0	57.4	65.3	73.5	81.6	91.3	90.2	82.2	66.5	55.3	45.3		
		MCWB	38.8	41.8	45.0	50.6	56.3	60.0	63.0	61.7	58.1	50.7	45.1	39.5		
	10%	DB	43.1	46.1	54.3	60.8	70.0	77.3	88.9	86.4	78.6	63.0	52.3	43.0		
		MCWB	37.1	39.4	44.0	47.7	54.3	58.1	62.3	60.5	56.6	48.9	43.5	37.2		
	0.4%	WB	44.0	47.6	51.2	57.4	63.0	66.7	68.2	66.0	64.0	57.7	52.2	45.4		
		MCDB	48.4	56.4	63.7	72.6	80.7	88.3	90.5	91.3	85.8	77.4	61.2	51.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.1	44.3	48.6	54.1	59.9	63.4	66.1	64.1	61.5	54.2	48.8	42.3		
		MCDB	45.9	51.8	60.3	69.1	75.7	82.6	89.3	89.1	83.2	69.4	58.9	47.4		
	5%	WB	39.4	42.1	46.5	51.2	57.4	61.0	64.4	62.6	59.4	51.8	46.1	40.2		
		MCDB	44.2	48.4	56.8	63.3	72.1	79.3	88.2	86.9	79.5	64.9	55.3	45.0		
	10%	WB	37.7	40.2	44.4	48.7	55.2	59.0	63.0	61.0	57.2	49.6	43.8	37.8		
		MCDB	42.5	45.9	53.3	59.3	67.6	76.0	86.6	84.2	76.4	61.3	51.7	42.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	14.3	19.8	22.1	23.6	26.2	34.1	34.7	32.3	24.8	15.6	11.3		
		MCDBR	12.6	18.1	26.8	33.5	32.8	34.6	39.3	41.4	41.4	34.1	21.6	12.1		
		MCWBR	8.9	11.6	16.0	19.0	17.0	16.7	16.6	17.5	19.7	18.7	13.1	8.7		
		MCDBR	11.8	16.7	24.6	30.5	29.3	32.0	36.3	37.5	37.1	30.6	19.8	11.7		
	5% WB	MCWBR	8.8	11.1	15.2	17.6	15.4	15.9	16.0	16.1	17.8	17.4	12.4	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.291	0.308	0.369	0.374	0.358	0.359	0.363	0.350	0.315	0.295	0.282		
	taud		2.375	2.465	2.485	2.297	2.328	2.405	2.422	2.403	2.417	2.521	2.513	2.462		
	Ebn at Noon		258	286	295	282	284	288	286	281	276	273	260	254		
	Edh at Noon		24	26	30	39	39	36	35	35	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		442	729	1102	1585	1913	2227	2427	2061	1532	951	519	348		
	RadStd		51	86	110	124	163	136	97	115	104	78	43	45		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page